COCO DETR Benchmark

Hungarian vs PM3 Scaling Study

April 10, 2026

Benchmark Design

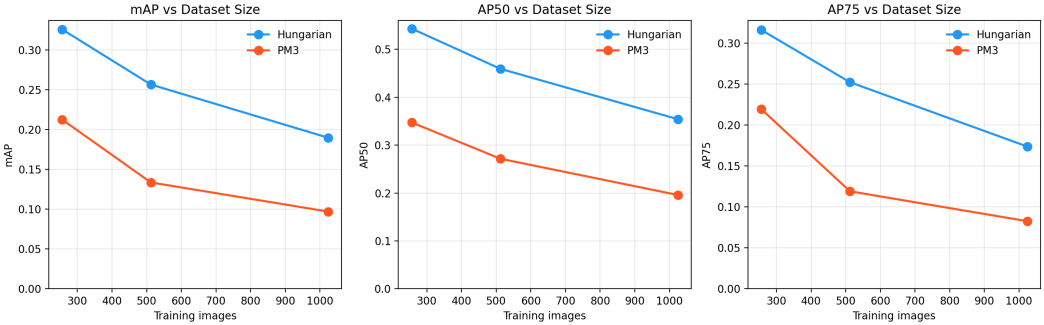
- Goal: compare Hungarian and PM3 matching in a vanilla DETR detection setting, varying only the assignment rule.
- Backbone: pretrained `facebook/detr-resnet-50`, fine-tuned end to end.
- Shared cost matrix: classification + box L1 ($\times 5$) + GIoU ($\times 2$).
- PM3 config: temperature $\tau = 0.12$, power $p = 3$.
- Training: AdamW, lr $= 10^{-5}$, weight decay 10^{-4}, gradient clip 0.1, 6 epochs, batch size 2.
- Evaluation: torchmetrics `MeanAveragePrecision` (COCO-style mAP).

Dataset Scaling Protocol

Scale	Train images	Val images	Purpose
Smoke	128	64	1-epoch sanity check
Small	256	128	Initial comparison
Medium	512	256	Scaling trend
Large	1024	512	Scaling trend

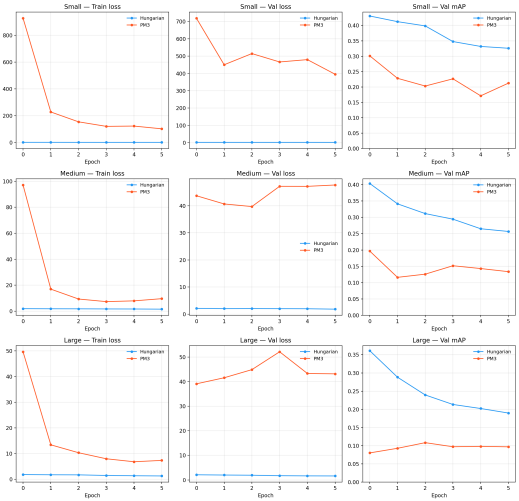
All subsets are drawn from real COCO 2017 (first N images by index). Same seed (42) and hyperparameters across all scales.

Scaling Comparison: mAP vs Dataset Size



Hungarian and PM3 detection quality as training data increases.

Training Curves by Scale

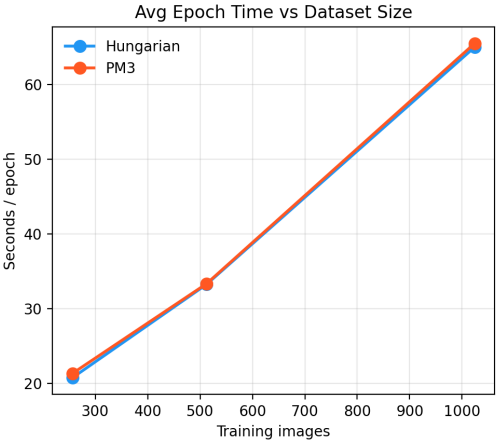


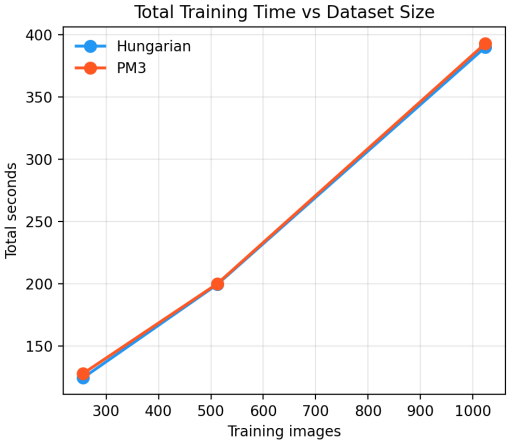
Key Numbers

Run	Train imgs	Val loss	mAP	AP50	AP75	Time (s)
Hungarian small	256	2.12	**0.326**	**0.543**	**0.316**	124
PM3 small	256	395.3	0.213	0.347	0.219	128
Hungarian medium	512	1.83	**0.257**	**0.459**	**0.252**	199
PM3 medium	512	47.7	0.134	0.272	0.119	200
Hungarian large	1024	1.62	**0.190**	**0.354**	**0.174**	390
PM3 large	1024	43.2	0.097	0.196	0.083	393

All runs: 6 epochs, batch size 2, lr $= 10^{-5}$, seed 42.

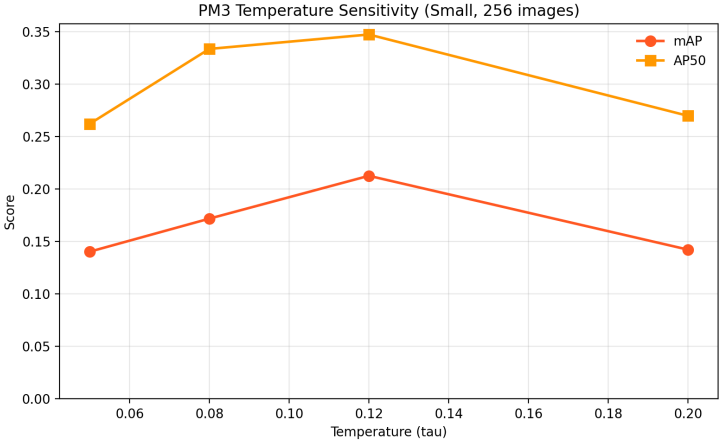
Wall-Clock Timing





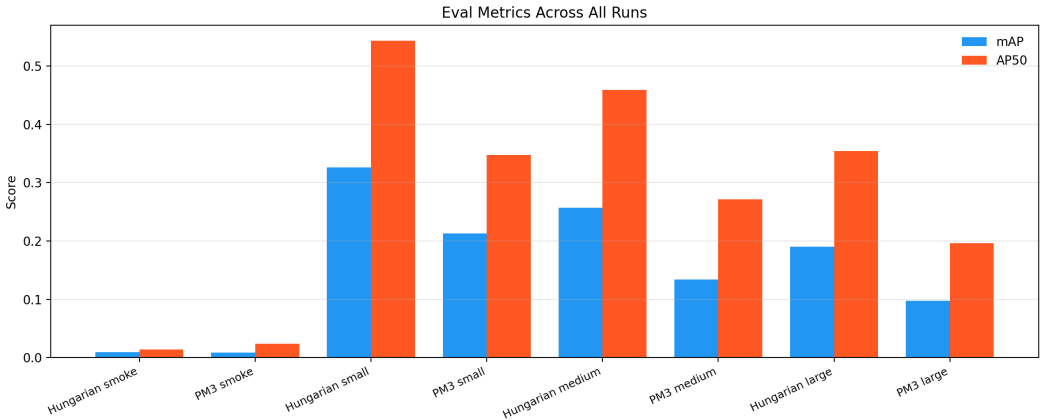
PM3 avoids the $O(N^3)$ Hungarian solver per image. At these small scales the overhead is dominated by forward/backward; timing differences emerge at larger N.

PM3 Temperature Ablation



τ	mAP	AP50
0.05	0.140	0.262
0.08	0.172	0.334
0.12	**0.213**	**0.347**

Eval Metrics Across All Runs



Observations

- Both matchers train successfully at all scales. Wall-clock time is nearly identical.
- **mAP decreases with more training data** for both matchers. This signals overfitting at 256 images and under-training at 512/1024 images with only 6 epochs.
- Hungarian leads PM3 at every scale, and the gap **widens** with more data: 0.11 at small, 0.12 at medium, 0.09 at large (relative gap grows from 35% to 49%).
- PM3 loss scale is 20–200$\times$ larger than Hungarian. The per-component $p = 3$ power amplification makes the loss landscape very different.
- Validation loss for PM3 drops sharply over epochs (from $>$700 to $\sim$43) but this doesn't translate to proportional mAP gains.

Next Steps

- **More epochs**: 6 is far too few. mAP *decreases* with more data because the model can't memorize larger datasets in 6 epochs. Try 20–50 epochs with early stopping.
- **Learning rate schedule**: Separate backbone lr ($10\times$ lower) and cosine/step decay. Standard DETR practice.
- ~~PM3 temperature sweep~~ Done: $\tau = 0.12$ confirmed near-optimal.
- **Loss normalization**: PM3 loss is 20–200$\times$ larger. Consider normalizing to make lr comparable across matchers.
- Scale to 5k+ images *after* fixing the training recipe — current results are dominated by under-training.
- Track per-category AP and duplicate prediction rate.